

FANUC Robot series

R-30*i*A, R-30*i*B, and R-30*i*B Plus CONTROLLER

***i*RVision[®] Error Handling**

Setup and Operations Manual

Applies to Version 7.70 and later

MAROCIERH02131E REV. B

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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

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FANUC America Corporation Patent List

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Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC Robotics is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC Robotics therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC Robotics systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC Robotics therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC Robotics training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC Robotics training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.
- Position all controllers outside the robot work envelope.

- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC Robotics recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
- Using a low motion speed, test run the program continuously for at least one full cycle.
- Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

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1

iRVISION ERROR HANDLING APPLICATION OVERVIEW

1 iRVISION ERROR HANDLING APPLICATION OVERVIEW

1.1 Overview

iRVision Error Handling is designed to provide a uniform method of recovery for TP programs that desire operator intervention in case of a vision error or part not found. Two macros have been created to replace the most common use of vision TP instructions.

Instead of calling the RUN_SNAP vision instruction, the vision GET_OFFSET, and testing for no parts, the user can replace these steps with one MACRO (Run_Find_Offset).

To use the macro the user supplies the Vision Process name, a status register and a vision register or result register for the Pass Fail result. If the instruction fails, Error Handling is executed. The operator must provide a response to continue. The default responses are Retry and Ignore.

The TP Programmer must provide coding to respond to the status register value. In general a status of 0 indicates success, a status of 1 indicates retry, and a status of 2 indicates ignore. By default the retry value is never returned.

Error Handling is designed to blend in with Spot Tool+ so that vision Error Handling looks and behaves like any other spot tool error. For non Spot Tool+ Systems, the procedure is roughly the same.

1.2 Features

- Fully integrated iRVision product
- Supports SpotTool+ using spot tool screens
- Supports non SpotTool+ systems using Material handling screens.
- Replaces RUN_SNAP and GET_OFFSET with Run_Find_Offset macro
- Replaces RUN_SNAP and GET_PASSFAIL with Run_Pass_Fail
- Warnings and/or errors are logged when vision encounters a vision error or part not found.
- Error Handling includes automatic retry. By default Error Handling makes five attempts before posting an error. By default, Error Handling waits 200 milliseconds between attempts. The first attempts are logged as warning. Warnings do not require operator intervention.
- The operator recovery method in response to an error is logged in the Alarm Log.
- Allows the user to customize up to six recovery options.

1.3 How Error Handling Works

The Error Handling macros replace the vision RUN_SNAP instruction and vision GET-OFFSET instruction with one macro call, Run_Find_Offset. The recovery option may be used in cases where a part must be found to continue production. An operator will be required to intervene in the case a part is not found or optionally, a test fails.

In this example, the Teach Pendant program calls a vision process and the part is not found. An alarm is posted reflecting the cause of the vision error. The error alarm causes the robot to fault. The program pauses and a recovery screen is displayed.

Operator intervention is required. The operator determines the cause of the fault. They may access the alarm log for more details. Once the cause of the alarm is determined, the operator selects a recovery option. The default options are Retry and Ignore. If Retry is selected the robot retries the vision process. If Ignore is selected the program continues. The Teach pendant code reacts to the value returned in the status register.

The TP programmer would normally program as follows:

```

1: VISION RUN_FIND 'BIGCIRCLE'      ;
2: VISION GET_OFFSET 'BIGCIRCLE' VR[1] JMP LBL[99] ;
3:L P[1] 2000mm/sec FINE VOFFSET,VR[1]      ;
4: END ;
5: LBL[99:Vision Error] ;
6: !Error Handling follows ;
7: . . .
8:

```

In the above example the programmer must still include coding for handing the response to the error. This may include writing menu screens or waiting for an I/O signal from a PLC screen.

Below is a coding example of the same process using Error Handling. The code is reduced to:

```

1: Run_Find_Offset('BIGCIRCLE',1,8) ;
2: IF R[8:status]<>0,JMP LBL[98] ;
3:L P[1] 2000mm/sec FINE VOFFSET,VR[1]      ;
4: END ;
5: LBL[98:No Offset] ;
6:L P[1] 2000mm/sec FINE      ;
7: END ;

```

The Menu recovery screens are part of the Error Handling option. It includes automatic retry. The user only needs to provide code for when the operator selects to ignore the error. In this case, the robot moves without a vision offset.

2

iRVISION ERROR HANDLING PRODUCT FAMILY

2 iRVISION ERROR HANDLING PRODUCT FAMILY

2.1 iRVision

iRVision is a visual sensor system designed to provide robot guidance and error proofing capabilities by inspecting a work piece using cameras. It can perform various measurements as well as adjust the robot motion so that the robot can manipulate the work piece in the same way as programmed. For full details on the setup and use of iRVision, please refer to the *iRVision Addendum* and *iRVision Operator* manuals.

3

KEY CONCEPTS AND DEFINITIONS

3 KEY CONCEPTS AND DEFINITIONS

3.1 Overview

The Vision Error Handling product is designed to provide an operator interface for vision errors. Errors and warnings are automatically displayed on the alarm log.

3.2 Error

A vision error is an unexpected occurrence that requires operator intervention. The event is log to the Alarm Log, A Recovery Menu is displayed, The system pauses.

The operator must detect the problem, select a recovery option, perform a FAULT RESET, and perform a CYCLE START.

3.3 Warning

A vision warning is an unexpected occurrence that warns the operator by logging to the alarm log. The Error Handling system will automatically retry up to the Maximum Retries threshold without operator intervention. The operator should check the log to see if the same warning occurs occasionally and determine if the vision process should be improved.

When the Maximum Retry threshold is reached, the warning is promoted to an error which will require operator intervention.

3.4 Maximum Retry

When an unexpected vision result occurs it is logged initially as a warning. Many times the problem may be part not found. This may be caused by a change in lighting. The system will retry rather than pause. This feature could reduce down time. The number of times a system will retry is defined by the Maximum Retry value. The default value is five. This means there will be four warning and one error posted for an error that repeats. This value can be reset by including it as the fifth argument in a call to `irErrHdl`

3.5 Retry Time

When a retry occurs automatically (Maximum Retry count not reached), Error Handling waits a defined number of milliseconds before retrying. The default time is 200 milliseconds. This value can be reset by including it as the sixth argument in a call to `irErrHdl`

3.6 No Part OK

This a Boolean value used to indicate if No Part present should be an error or an acceptable result. By default (Boolean is False) an error or warning is logged when no part is found. When this value is enabled and no part is found, the Status code is 7.

3.7 Post No Pass

This is a Boolean used to indicate if an Error Proofing vision Process failure should log an alarm. By default, (Boolean is True) it will not post an alarm on a failure.

4

iRVISION ERROR HANDLER FEATURES

4 iRVISION ERROR HANDLER FEATURES

The iRVision Error Handling software provides many features for the user.

- Simple interface for Error Handling that blends in with current method (for SpotTool+)
- Reduce programming by providing recovery interface.
- Ability to automatically retry marginal vision error
- Ability to Allow No Part as an error or an acceptable result.
- Ability to require / Not Require operator intervention for Error Proofing.
- Logs all warning and Errors to Alarm Log.
- Logs operator recovery option selection to Alarm log.
- Provides extended Menu Recovery Options in line with argument list. For example this call has four valid options, CALL
`irErrHdl('SnapFind','BigCircle',1,8,'RETRY','IGNORE','ABORT','NOMINAL')`
- High use functions have defined macros. Run_Find_Offset and Run_Pass_Fail

5

HARDWARE SETUP

5 HARDWARE SETUP

5.1 Overview

There is no hardware setup for the error handling option.

6 SOFTWARE SETUP

6.1 Overview

To setup the Error Handling software, the following steps are required:

- Complete Hardware Setup (see previous chapter)
- Install robot option.
- Install the robot macros in the system macro table.

6.2 Install Error Handling Software

The required Robot Options is Error Handling Software (Robot option R757).

After installation is complete the following programs will be installed

KAREL programs (These KAREL programs are not visible)

- Iredata - data area for error handling
- Irerecsc - task that communicates with material handling version of forms
- Irrrhdl - task that coordinates vision calls and error logging

TP programs

- Iretest.tp - Set of example code for using with Handling Tool
- Iretestspot.tp - Set of example code for using with SpotTool+
- Runoffst.tp - TP program that supports the RUN_GET_OFFSET macro
- Runpassf.tp - TP program that supports the RUN_PASS_FAIL macro

Form Dictionary

- Errhd - Form dictionary for Handling Tool recovery screens

The Error Handling option requires either the iRVision 2D package or the iRVision 3D Package. The Error Handling option includes KAREL programs (invisible to the user), TP coding examples of programs for calling vision, Vision Data files, and form dictionary.

6.3 System Macro Table Setup

If the TP programmer desires to use macros, then they must be setup in the System Macro Table. The user can define the macro as shown in the figure below manually or use the example in iretest.tp.

For the manual method refer to the *Handling Tool Setup and Operations Manual* or the *SpotTool+ Setup and Operations Manual*, in the Macro Commands section of the General Setup Chapter.

The automatic method is to open iretest.tp, cursor to line 71, and run the program from this point. The code is setup to start placing macros at entry 10. The user can edit line 73 and modify the third parameter to a different starting table entry. Note the user may want to use a different vision status register (register R[5]).

Figure 6-1 Coding Snippet from IRETEST.TP for setting up SYSTEM MACRO table.

```

71:  LBL[15]
72:  R[5]=(-1)
73:  CALL IRERRHDL('InitMacros',0,10,
   :  5)

```

Figure 6-2 System Macro Table after adding the Error Handling macros.

	Instruction name	Program	Assign
8	[	][	--[0]
9	[	][	--[0]
10	[Run_Find_Offset	][RUNOFFST]	--[0]
11	[Run_Pass_Fail	][RUNPASSF]	--[0]
12	[	][	--[0]
13	[	][	--[0]

Using the Run_Find_Offset and Run_Pass_Fail macros is entirely up to the TP Programmer. The programmer may alternatively use:

CALL IRERRHDL('SnapFind', ...) instead of Run_Find_Offset (...)

and

CALL IRERRHDL('SnapPassFail', ...) instead of Run_Pass_Fail (...)

7

CODING WITH *i*RVISION ERROR HANDLING

7 CODING WITH *i*RVISION ERROR HANDLING

7.1 Overview

Using Error Handling for the basic recovery methods is quite simple. A call is made to Error Handling with the RUN_FIND_OFFSET. The Error Handling process performs both the Snap and Find, and the GET Offset functions.

A vision status register indicates if the call was successful or the user elected to ignore the error (vision offset is not valid).

In the event a vision process fails, the process is retried automatically before requesting operator intervention. The operator recovers by retrying the process after making some adjustments or recommending that the vision result should be ignored.

7.2 Vision Status Register

The user includes the index of the vision status register in the call to Error Handling. The index value is set to indicate the results of the call. The same value is store internally in the KAREL program iredata in the INTEGER variable StatusResult. The table shows the results and their meaning.

Table A-1 Vision Status Values

Mnemoic	Value	Description
Ire_Pending	-1	(internal) Error Handling Incomplete
Ire_Success	0	Vision Operation Successful
Ire_retry	1	(internal) Error Handling is automatically retrying operation
Ire_Ignore	2	Operator has selected to ignore the vision result
Ire_User3	3	Operator has selected the third recovery option
Ire_User4	4	Operator has selected the fourth recovery option
Ire_User5	5	Operator has selected the fifth recovery option
Ire_User6	6	Operator has selected the sixth recovery option
Ire_NoParts	7	IF NoPartOK = TRUE AND no parts were found
Ire_Failed	8	IF PostNoPass = FALSE AND Error Proof VP Pass/Fail result register = 0.
Ire_NotDone	9	IF PostNoPass = FALSE AND Error Proof VP Pass/Fail result register = 2.
Ire_Repeat	10	(internal) Automatic retry issued retry snap and get offset
Ire_abort	11	(internal) Error Handling Aborted
Ire_Recall	12	(internal) Recovery aborted when screen exited, repost the error

The descriptions labeled “(internal)” are never returned to the TP program.

7.3 RUN_GET_OFFSET Macro

This macro performs the functionality of two vision instructions, Run_Find and Get_Offset. It also contains provisions for automatic retries, error logging, and operator recovery.

The *Run_Find_Offset* macro is used replace the vision RUN_FIND and GET_OFFSET instructions. When using the standard method it is assumed that production will wait until a part is found or the operator allows it to continue without a vision result.

Syntax:

Run_Find_Offset (VPname, VR, Status)

- VPname - String containing name of vision process
- VR - Integer that specifies the Vision Register to receive the results
- Status - Integer that specifies the general register to receive the result code.

CALL irErrHdl (FC, VPname, VR, Status)

- FC - String containing the function name or Integer containing the function code.
- VPname - String containing name of vision process
- VR - Integer that specifies the Vision Register to receive the results
- Status - Integer that specifies the general register to receive the result code.

Example:

In the next example the user performs the same functionality as above. It includes a recovery menu with Retry and Ignore Options. In the example below entering Ignore would use the current VR[1] result.

Example: Code using the Run_Find_Offset macro.

```

1: Run_Find_Offset ('BigCircle',1,5)
2: IF R[5:Status Result] = 2, JMP LBL[20]
3: L P[1] 100mm/sec Fine VOFFSET, VR[1]
4: END
5: LBL[20: No Offset]
6: L P[1] 100mm/sec Fine
7: END

```

Example: Code using the CALL irErrHdl routine.

Note that line 1 could be replaced with

1: CALL irErrHdl('SnapFind','BigCircle',1,5)

or

1: CALL irErrHdl(3,'BigCircle',1,5)

and produce the same results.

In the default mode, the vision status return code will always be:

- 0 - indicating success
- 2 - indicating ignore
- 8 - Failed inspection
- 9 - Test inconclusive (parent part not found). Not this only occurs when a parent tool is used to locate a tool that performs the test. Example. The parent tool, GPM finds a hole and the child tool, EDGE_PAIR measures the diameter. If the hole is not found the diameter can not be measured.

If the user does not accept failures or inconclusive tests, PostNoPass = TRUE. then the vision status return code will be:

- 0 - indicating success
- 2 - indicating ignore

To require that a part must pass, the user must set the control Boolean, PostNoPass to FALSE. When set to FALSE, any failure will require operator intervention. The user can set this Boolean variable FALSE using:

```
CALL irErrHdl('SetVar','PostNoPass','FALSE')
```

The Boolean remains TRUE until the variable is set to TRUE using:

```
CALL irErrHdl('SetVar','PostNoPass','TRUE')
```

TRUE is the default value. Note: Rebooting the controller does not reset the value.

7.4 RUN_PASS_FAIL Macro

This macro performs the functionality of two vision instructions, Run_Find and Get_PASSFAIL. It also contains provisions for automatic retries, error logging, and operator recovery.

The *Run_Pass_Failt* macro is used replace the vision RUN_FIND and GET_PASSFAIL instructions. When using the standard method it is assumed that production will wait until the Get_Pass_Fail returns a pass fail result or the operator allows it to continue without a pass fail result.

Syntax:

Run_Pass_Fail (VPname, PResult, Status)

VPname - String containing name of vision process

- | | |
|----------|---|
| PFresult | - Integer that specifies the General Register to receive the results |
| Status | - Integer that specifies the General Register to receive the result code. |

CALL irErrHdl (FC, VPname, VR, Status)

- | | |
|----------|--|
| FC | - String containing the function name or Integer containing the function code. |
| VPname | - String containing name of vision process |
| PFresult | - Integer that specifies the General Register to receive the results |
| Status | - Integer that specifies the general register to receive the result code. |

Example

In the next example the user performs the same functionality as above. It includes a recovery menu with Retry and Ignore Options. In the example below entering Ignore would use the current VR[1] result.

Example: Code using the Run_Pass_Fail macro

```

1: Run_Find_Offset ('TestSize',8,5)
2: IF R[5:Status Result] = 2, JMP LBL[20]
3: IF R[8] = 1, JMP LBL[100]
4: IF R[8] = 0, JMP LBL[900]
5: IF R[8] = 2, JMP LBL[800]
6: END
7: LBL[100: Passed]
8: END
9: LBL[900: Failed]
10: END
11: LBL[800: Inconclusive]
12: END

```

Example: Code using the CALL irErrHdl routine

Note that line 1 could be replaced with

```
1: CALL irErrHdl('SnapPassFail','TestSize',8,5)
```

or

1: CALL irErrHdl(9,'TestSize',8,5)

and produce the same results.

In the default mode, the vision status return code will always be:

- 0 - indicating success
- 2 - indicating ignore

If the user has allowed for No Part Found to be an acceptable result, then the vision status return code will be:

- 0 - indicating success
- 2 - indicating ignore
- 7 - No part found.

To allow the No Part Found return code (7), the user must set the control Boolean, NoPartOk to TRUE. The user can set this Boolean variable TRUE using:

CALL irErrHdl('SetVar','NoPartOK','TRUE')

The Boolean remains TRUE until the variable is set to FALSE using:

CALL irErrHdl('SetVar','NoPartOK','FALSE')

FALSE is the default value. Note Rebooting the controller does not reset the value.

7.5 Entering the Vision Process Name.

The controller software has been modified to present vision process names when entering a string value.

A new method has been added to the controller for selecting string names that are defined as vision processes. Note that only vision processes are included. That is there is no provision for camera names or camera calibration names.

The new method will assist the programmer by making string entry easier for vision process names.

The Macros and calls to irErrHdl require adding arguments. Figure 7-1 Editing the Macro shows the macro in the Edit screen before the vision name parameter has been added.

Figure 7-1 Editing the Macro

TEST

1: RUN_FIND_OFFSET 1/2

[End]

			[CHOICE]		
--	--	--	----------	--	--

When you select F4, the [CHOICE] key, the drop down list shown in Figure 7-2 Adding a String Parameter is presented. The user should select the third item, “String”.

Figure 7-2 Adding a String Parameter

The screenshot shows a window titled "TEST" with a dark blue header. The main area is white and contains a code editor with the following text:

```
1: RUN_FIND_OFFSET  
[End]
```

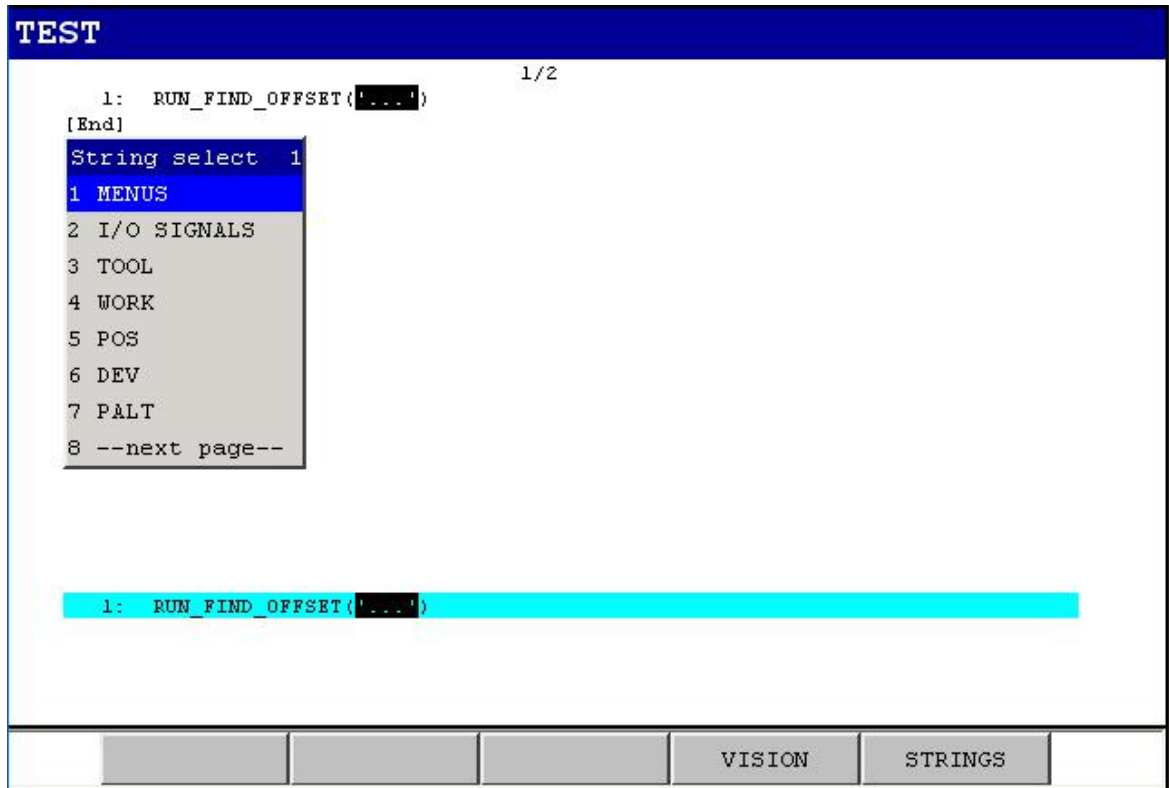
To the right of the code editor, the text "1/2" is displayed. A dropdown menu is open, showing a list of options:

- Parameter select 1
- 1 R[]
- 2 Constant
- 3 String
- 4 &R[]
- 5 <None>
- 6 <Insert>
- 7 SR[]
- 8

The option "3 String" is highlighted in blue. Below the code editor, there is a red horizontal bar with the text "1: RUN_FIND_OFFSET". At the bottom of the window, there is a toolbar with several buttons, including one labeled "[CHOICE]".

The function keys appear. The F4, [VISION] key will appear when there is a list of vision process names defined on the system. F5, [STRINGS] key will always appear.

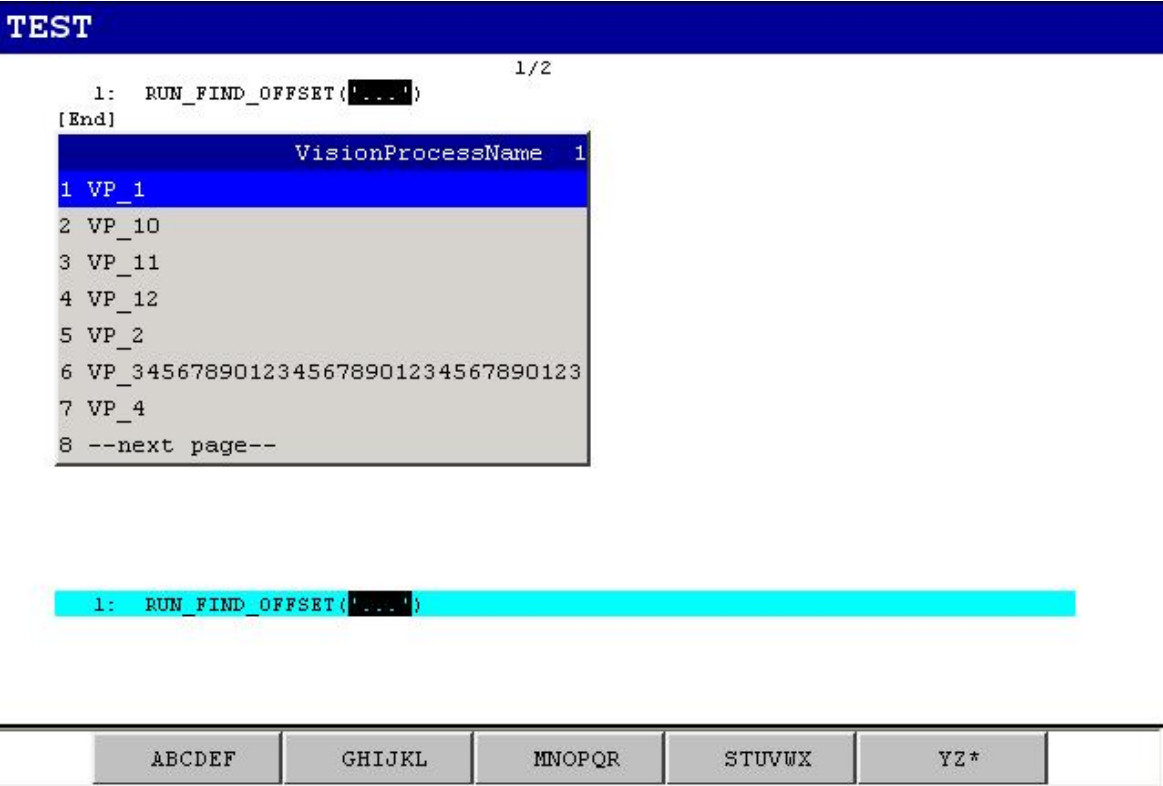
Figure 7-3 New F4 [VISION] key



Pressing F4, the [VISION] key displays a drop down list of Vision Processes as shown in Figure 7-4 Select a vision process name from the list.

The programmer can select a vision process name.

Figure 7-4 Select a vision process name from the list



7.6 Customized Functions

7.6.1 Overview

iRVision Error Handling has defined macros for the two most common functions:

- RUN_GET_OFFSET (VPname, VR, Status)
- RUN_PASS_FAIL (VPname, PassFailReg, Status)

These macros in turn pass their parameters to these TP program:

- RUNOFFST.TP (VPname, VR, Status)
- RUNPASSF.TP (VPname, PassFailReg, Status)

These TP programs in turn call the KAREL program with different function codes and parameters:

- irErrHdl ('SnapFind', VPname, VR, Status)
- irErrHdl ('SnapPassFail', VPname, PassFailReg, Status)

As one can see, all TP calls and macros calls are just methods of calling irErrHdl with the right arguments and function code.

iRVision Error Handling supports these KAREL vision built-ins. See the *KAREL Reference Manual* for more information

Supported KAREL Built-Ins:

- V_CAM_CALIB (cal_name, func_code, status)
- V_GET_OFFSET(vp_name, register_no, status)
- V_GET_PASSFL (vp_name, register_no, status)
- V_RUN_FIND(vp_name, camera_view, status)
- V_SET_REF(vp_name, status)

IrErrHdl has defined a function code for each of these function and combinations functions. The user can easily add macros of heir own if they so desire.

7.6.2 Calling irErrHdl

The use of the argument list is defined by the function code, the first argument.

Normal Syntax:

CALL irErrHdl (FC, DataName, Result Register, Status Register, ...)

FC - String containing the function name or Integer containing the function code.

- Data Name - String containing name of vision process or camera calibration
- Result register - Integer that specifies the register to receive the results
- Status register - Integer that specifies the register to receive the status code.

Extended Syntax:

CALL irErrHdl (FC, DataName, Result Register, Status Register, Maximum Retries, Retry Time)

- FC - String containing the function name or Integer containing the function code.
- Data Name - String containing name of vision process or camera calibration
- Result register - Integer that specifies the register to receive the results
- Status register - Integer that specifies the register to receive the status code.
- Max Retries - Number of time irErrHdl retries warning before posting an error (default is 5)
- Retry Time - Wait time between retries (in milliseconds). Default is 200

Super Extended Syntax:

CALL irErrHdl (FC, DataName, Result Register, Status Register,

Menu1, Menu2, Menu3, Menu4, Menu5, Menu6)

- FC - String containing the function name or Integer containing the function code.
- Data Name - String containing name of vision process or camera calibration
- Result register - Integer that specifies the register to receive the results
- Status register - Integer that specifies the register to receive the status code.
- Menu n - Menu n contains a recovery option. It is a maximum of 32 characters. In the event of an error these options are presented in the recovery screen.

Menu 1 is a special case. It always refers to retrying the vision functions. That is Menu 1, is never returned to the user. If menu 1 is an integer, then it is assumed to be a max retry value and the default, 'RETRY' is option 1.

If menu 2 is an integer, then it is assume to be the Retry Time. F it is a string than it is recovery option 2. Menu items 3 through 6 must be strings. What ever selection is made by the operator during recovery except 1 is returned in the vision status register.(2 – 6)

Menu options are not saved in memory. They must be reset for each call.

See A.1.1..1.1B for a complete of all function codes and their parameter list.

7.7 TP Example Programs

Two TP Example programs have been supplied for familiarization with the *iR*Vision Error Handling capabilities.

- IreTest.tp – Used to test Handling Tool
- IreTestSpot.Tp – Used to test SpotTool+

Both programs work with a set of vision data files. The user should create the following vision programs.

- Camera1 - Camera Setup
- Grid15mm - Camera Calibration Setup – Grid calibration method 1 plane.
- BigCircle - 2D Single Vision Process that contains a GPM tool to find the large circles of the calibration grid.
- TestSize - Error Proofing Vision Process that contains a GPM to find the large or small circle of the Calibration grid and an Edge Pair location tool to measure the diameter of the circle.

By moving the grid or changing the search area, the user can create errors such as part not found for the BigCircle VP and create test failures for TestSize VP.

8

iRVISION ERROR HANDLING INTERFACE

8 iRVISION ERROR HANDLING INTERFACE

8.1 Overview

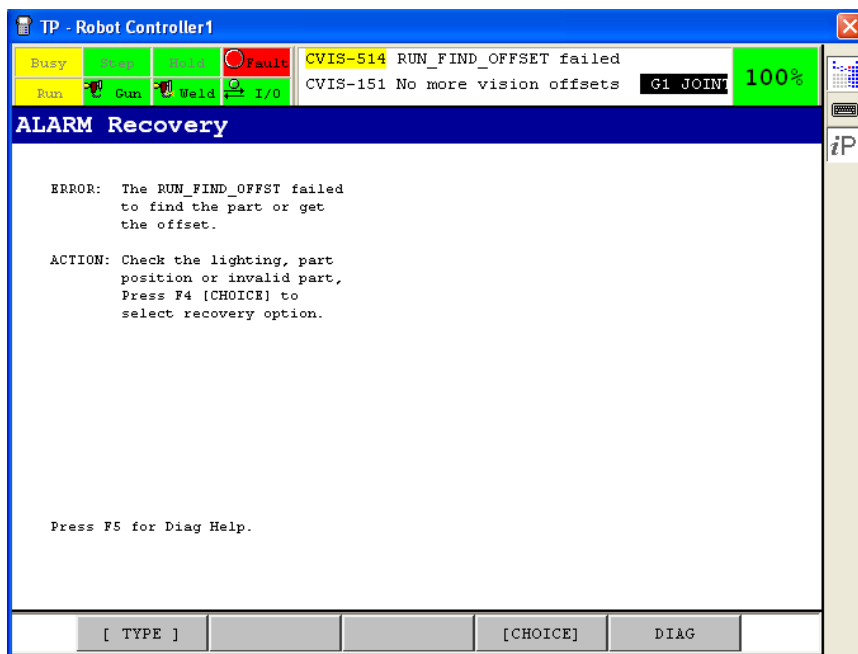
The Error Handling option is supported by multiple controller platforms. There are two methods of handling recovery. One method supports SpotTool+, the other method supports all other platforms.

8.1.1 SpotTool+ Option

If the system has the robot option H590, SpotTool+ loaded, it is considered a SpotTool+ system and the recovery options are designed to blend with other Spot Tool Recovery methods.

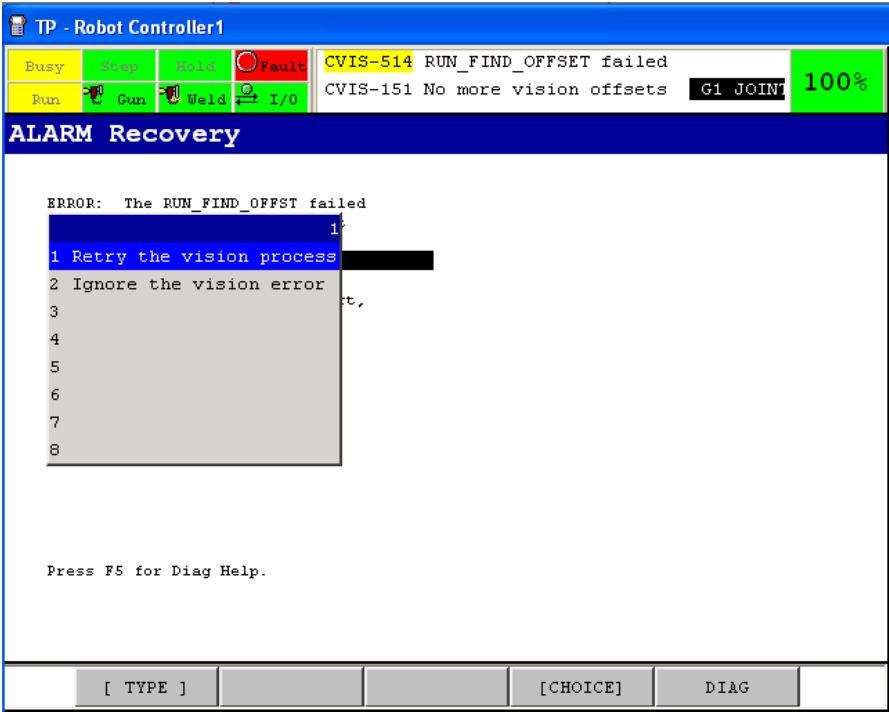
When a vision error occurs, the following screen is posted.

Figure 8-1 SpotTool+ Alarm Recovery screen



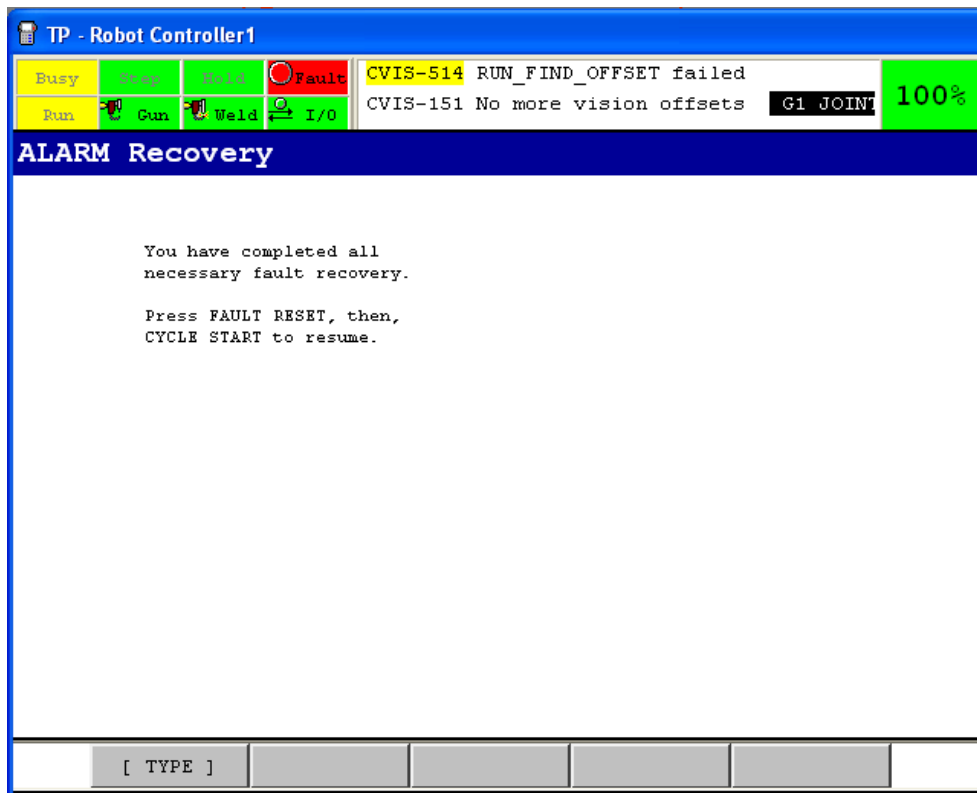
The user should determine the problem, resolve the problem, select the [Choice] key, and select an option from the pop up menu.

Figure 8-2 Error Handling Pop Up Options



Once an option is select another screen appears.

Figure 8-3 Alarm Recovery screen after recovery selection



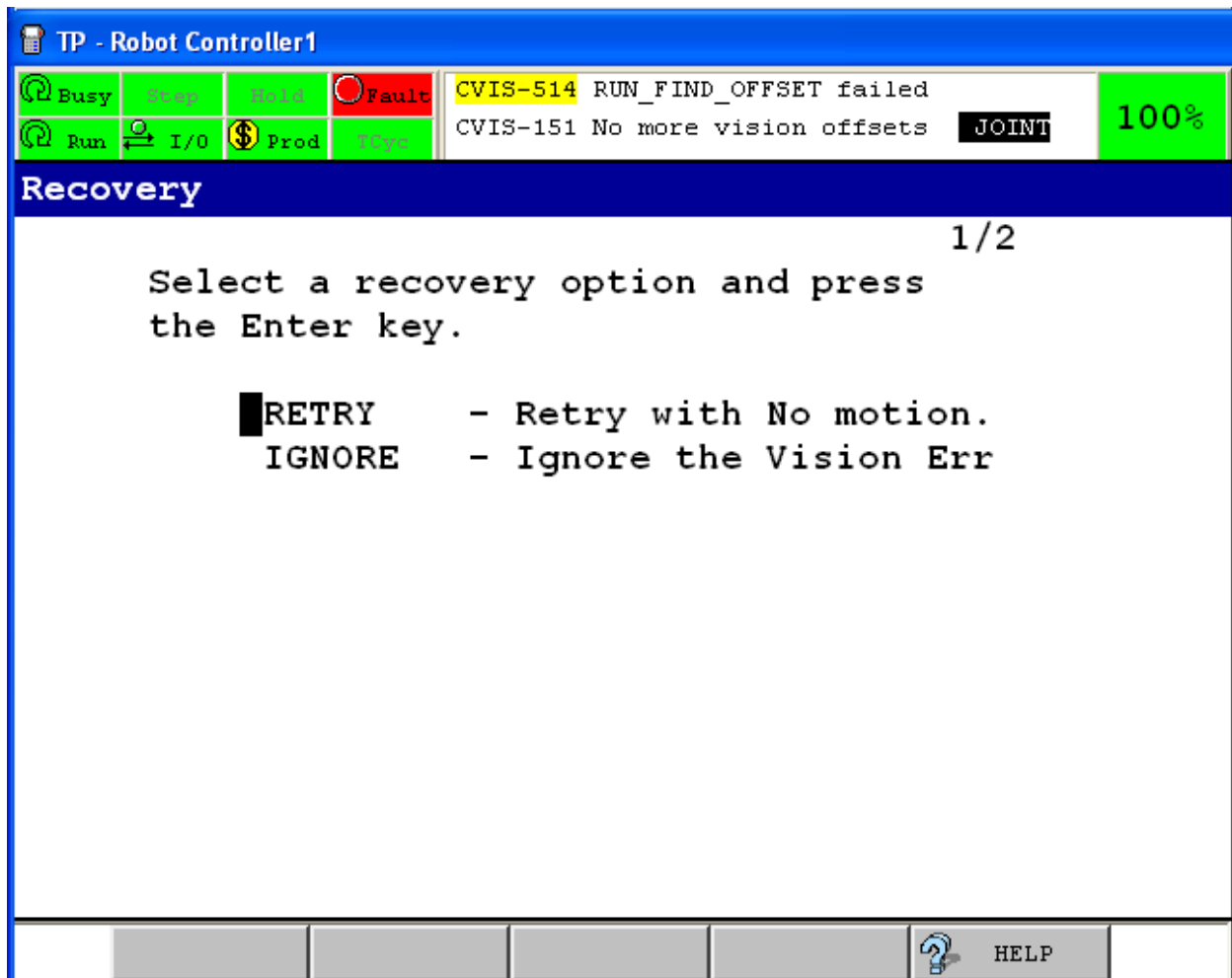
Follow the instructions, Press FAULT RESET and then CYCLE START.

8.1.2 Handling Tool Operating Systems (non SpotTool+)

If the system does not have the robot option H590, SpotTool+ loaded, it is considered a Handling Tool like system and the recovery screens use the standard form.

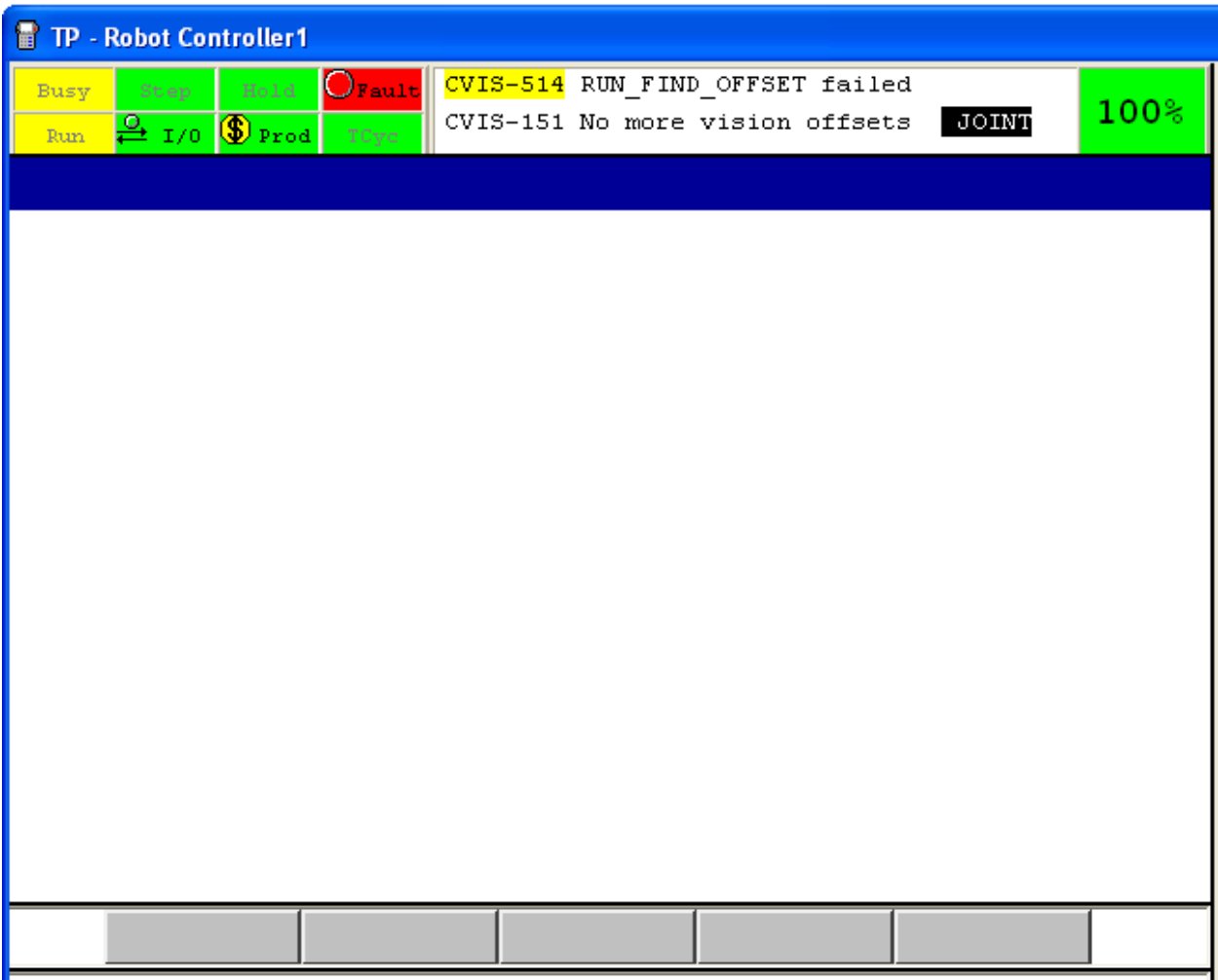
When a vision error occurs, the following screen is posted.

Figure 8-4 Handling Tool Recovery screen



The user should determine the problem, resolve the problem, and select the recovery option. A blank screen appears.

Figure 8-5 Error Handling after selecting an option



The user should RESET FAULT and CYLCE START. The screen before the error posting is refreshed.

9

SAVING AND RESTORING

9 SAVING AND RESTORING

9.1 TP Programs

Teach Pendant programs can be saved using the normal method found in the operator's manual. The list of TP programs used with IREERRHDL is listed below.

These programs are examples that the user can copy for their application.

- itetest.tp
- iretestspot.tp
- runpassf.tp
- runoffst.tp

9.2 KAREL Data

All KAREL data is stored in the KAREL program, IREDATA. The VR data can be saved to unique file names. Use KCL to save a VR file

Syntax: SAVE VARS < **file_name** > <= **prog_name** >

Example: SAVE VARS iredata1 = iredata

See the *KAREL Reference Manual's* Appendix – “KCL Command Alphabetical Description” for complete details. The user can also write a KAREL program to perform the KCL command.

9.3 Vision Data

Vision data files are stored on the *FR:* device. The file can be shown on the Teach Pendant using the Menu Files command. Set the device to *FR:* and use the directory command for *.vd.

Copy the file to an MC: or UD1 device.

A

ERROR CODES

A ERROR CODES

This section describes the error codes and their cause.

Table A-1

Error	Severity	Error Code	Description
<i>VXE_E_RUNOF</i>	PAUSE	<i>117 514</i>	<i>Run Get OFFSET failed</i>
<i>VXE_E_EP_FA</i>	PAUSE	<i>117 517</i>	<i>Run PassFail failed</i>
<i>VXE_E_EP_IN</i>	PAUSE	<i>117518</i>	<i>Run Pass Fail inconclusive</i>
<i>VXE_E_OPR_C</i>	WARN	<i>117519</i>	<i>Operator selected Custom Recovery</i>
<i>VXE_E_OPR_R</i>	WARN	<i>117520</i>	<i>Operator selected Retry Recovery</i>
<i>VXE_E_OPR_I</i>	WARN	<i>117521</i>	<i>Operator selected Ignore Recovery</i>

B

PROGRAMMING AND CUSTOMIZING IRVISION ERROR HANDLING

B PROGRAMMING AND CUSTOMIZING *i*RVISION ERROR HANDLING

The Error Handling software supports multiple functions. The KAREL program `irErrHdl` can be called by a Teach Pendant program. This is the runtime interface. The user can write programs to call the various KAREL vision functions, and set and clear control variables. The table below displays the supported functions

B.1 Calling a Function

Call `IRERRHDL` using

`CALL irErrHdl(FC, VP, VR, Status Index, Max Retries, Retry Time)`

FC	-- Function Code
VP	-- String Vision Process Name
VR index	-- Index of Vision Register
Status Index	-- Register to receive status (0 is success)
Max Retries	-- Number of auto Retries before positing an error
Retry Time	-- Wait interval between retries

B.1.1 KAREL Call

This method is the easiest to implement, but it requires knowing the function codes. The first parameter of the KAREL call is always the function code. The user can use a integer or String as the function code. The remaining arguments may be optional depending on the function.

B.1.2 TP Call

This method has the name of the function encrypted in the name of the TP program. It has one less argument, the function code is missing. The advantage of this method is the comments in the TP program describe the arguments and make it easier to remember. The only disadvantage is requiring an extra program. The user could have used the KAREL call method and then the TP program would not need to be on the robot. Only two macros are provided with the software installation, `RUN_GET_OFFSET`, and `RUN_PASS_FAIL`. The user can easily add macros and/or TP programs.

B.2 Function Description and Argument List Table

B.2.3 irErrHdl Functions

The table below indicates which arguments are valid for a function.

Table B-1

Arg 1	Arg1	Arg2	Arg3	Arg4	Arg5	Arg6
Name (String function)	Function Value	Vis Proc	Vision Reg	Status Reg	Max Retries	Retry Time
Snap	1	VP	x	O	O	O
Find	2	VP	VR	O	O	O
SnapFind	3	VP	VR	O	O	O
SetRef	4	VP	x	O	O	O
SnapSetRef	5	VP	x	O	O	O
Calibrate1	6	C	x	O	O	O
Calibrate2	7	C	x	O	O	O
PassFail	8	VP	R	O	O	O
SnapPassFail	9	VP	R	O	O	O
Snap1	10	VP	V	O	O	O
Snap2	11	VP	x	O	O	O
Snap3	12	VP	x	O	O	O
Snap4	13	VP	x	O	O	O
Help	14	x	x	x	x	x
InitMacros	15	x	E	x	x	x
SetVar	16	VN	V	O	O	O

O indicates optional

VP indicates Visoin Process name required

C indicates Calibration Name required.

VN indicates Variable Name required

E indicates entry (Integer) required

V indicates string value required

X indicates the argument is ignored

Note in many cases the Register Index is a direct value see examples

B.2.4 Extended Argument List

With the Handling Tool version pf IrErrHdl, the user can customize the operator response with up to six responses. Normally Error Handling allows the operator to enter Retry or Ignore. Using additional arguments the user can redefine the acceptable responses.

CALL irErrHdl(FC, VP, VR, Status Index, Max Retries, Retry Time, Menu3, Menu4, Menu5, Menu6)

FC	-- Integer or string containing Function Code
VP	-- String Vision Process Name
VR index	-- Integer containing Index of Vision Register
Status Index	-- Integer containing Register to receive status (0 is success)
Max Retries/Menu1	-- If Integer then Number of auto Retries before positing an error
	Or
	-- If String then it contains first operator response option (redefined)
Retry Time/Menu2	-- If integer, then Wait interval between retries
	Or
	-- If String, then it contains the second operator response option
Menu3	-- String containing third operator response option
Menu4	-- String containing fourth operator response option
Menu5	-- String containing fifth operator response option
Menu6	-- String containing sixth operator response option

Note the Menu Items can have a length of 32 characters. The first eight characters should begin with a letter and have trailing blanks to fill . The ninth character should be a dash. The tenth character should be a blank. The remaining characters can be any letter.

The TP programmer must test the vision status register and react to the return code.

Example 1:

```
CALL irErrHdl ('SNAPFIND', 'BigCircle,1,5,0,0,
'Abort - Abort process after going home',
'Last - Use the last vision offset')
```

Result 1:

The user retains the Retry and Ignore options as 1 and 2 (default) and adds Abort and Last as option 3 and 4. If the operator selected Abort, then the status code in R[5] would be a 3. If the operator selected Last, then the status code in R[5] would be a 4. Note there are only four options, because the ninth and tenth parameter were omitted.

Note that Option 1 function is always a retry. The user may rename it to 'Retest', 'Repeat', 'Try Again'. Etc...

Example 2:

```
CALL irErrHdl ('SNAPFIND', 'BigCircle,1,5,'Retest - Retest vision',0,
'Abort - Abort process after going home',
'Last - Use the last vision offset',
'Nominal - Use nominal position')
```

Result 2:

The user retains the Ignore options as 2 (default), redefines Retry to be Retest, and adds Abort, Last, and Nominal as option 3, 4, and 5. If the operator selected Abort, then the status code in R[5] would be a 3. If the operator selected Last, then the status code in R[5] would be a 4. Note there are only five options, because the tenth parameter was omitted.

Note that Option 1 function is always a retry. The user may rename it to 'Retest', 'Repeat', 'Try Again'. Etc...

B.2.5 Parameter List Examples

The parameters are used differently for each function.

- See the *iretest.tp* program for examples of how the functions are used with a Handling Tool platform.
- See *iretestspot.tp* program for or examples of how the functions are used with SpotTool+ platform

B.3 IRERRHDL Functions

B.3.1 SNAP

This function requests IRERRHDL to Snap an image.

Example:

R[5:status] = 0

CALL IRERRHDL ('SNAP' 'BigCircle', 0,5)

R[5] is zero if successful

'BigCircle' is the vision process name

'Snap' is the function code

B.3.2 FIND

This function requests IRERRHDL to perform a get offset function.

Example:

R[5:status] = 0

CALL IRERRHDL('Find' 'Bigcircle', 1,5)

VR[1] is the vision register

R[5] is zero if successful

'BigCircle' is the vision process name

'Find' is the function code

B.3.3 SNAPFIND

This function requests IRERRHDL to snap and then perform a get offset function.

Example:

```
R[5:status] = 0
CALL IRERRHDL('SnapFind' 'Bigcircle', 1,5 )
```

VR[1] is the vision register
R[5] is zero if successful
'BigCircle' is the vision process name
'SnapFind' is the function code

B.3.4 SETREF

This function requests IRERRHDL to set the reference position after a Run Find has been performed

Example:

```
R[5:status] = 0
CALL IRERRHDL('SetRef' 'Bigcircle', 0,5 )
```

R[5] is zero if successful
'BigCircle' is the vision process name
'SetRef' is the function code

B.3.5 SNAPSETREF

This function requests IRERRHDL to perform a Run Find followed by the set the reference position function.

Example:

```
R[5:status] = 0
CALL IRERRHDL('SnapSetRef' 'Bigcircle', 0,5 )
```

R[5] is zero if successful
'BigCircle' is the vision process name
'SnapSetRef' is the function code

B.3.6 Calibrate1

This function requests IRERRHDL to perform a calibration of the first plane..

Example:

```
CALL IRERRHDL('Calibrate1','Grid15mm',1)
```

'Calibrate1' is the function code

'Grid15mm' is the camera calibration setup name.

1 is the plane (request number)

B.3.7 Calibrate2

This function requests IRERRHDL to perform a calibration of the second plane..

Example:

```
CALL IRERRHDL('Calibrate1','Grid15mm',2)
```

'Calibrate1' is the function code

'Grid15mm' is the camera calibration setup name.

2 is the plane (request number)

B.3.8 PassFail

This function requests IRERRHDL to perform a Get_PassFail instruction. The user must first perform the Snap function.

Example:

```
R[5:status] = 0
```

```
CALL IRERRHDL('PassFail','TestSize',7,5)
```

'PassFail' is the function code

'TestSize' is the error proofing Vision Process.

R[7] Contains the pass fail result Failed, Passed, Inconclusive. (0,1, or 2)

B.3.9 SnapPassFail

This function requests IRERRHDL to perform a Snap function followed by the Get_PassFail instruction.

Example:

```
R[5:status] = 0
```

```
CALL IRERRHDL('SnapPassFail','TestSize',7,5)
```

'PassFail' is the function code

'TestSize' is the error proofing Vision Process.

R[7] Contains the pass fail result Failed, Passed, Inconclusive. (0,1, or 2)

R[5] is the status register

B.3.10 Snap1

This function requests IRERRHDL to perform a Snap function for view 1. It is intended to be used with a multi view vision process to find the target in the first view.

Example:

R[5:status] = 0

CALL IRERRHDL('SnapPassFail','BigCircle',0,5)

'SnapPassFail' is the function code

'BigCircle' is the Vision Process.

R[5] is the status register

B.3.11 Snap2

This function requests IRERRHDL to perform a Snap function for view 2. . It is intended to be used with a multi view vision process to find the target in the second view.

Example:

R[5:status] = 0

CALL IRERRHDL('SnapPassFail','BigCircle',0,5)

'SnapPassFail' is the function code

'BigCircle' is the Vision Process.

R[5] is the status register

B.3.12 Snap3

This function requests IRERRHDL to perform a Snap function for view 3. . It is intended to be used with a multi view vision process to find the target in the third view.

Example:

R[5:status] = 0

CALL IRERRHDL('SnapPassFail','BigCircle',0,5)

'SnapPassFail' is the function code

'BigCircle' is the Vision Process.

R[5] is the status register

B.3.13 Snap4

This function requests IRERRHDL to perform a Snap function for view 4. It is intended to be used with a multi view vision process to find the target in the fourth view.

Example:

R[5:status] = 0

CALL IRERRHDL('SnapPassFail','BigCircle',0,5)

'SnapPassFail' is the function code

'BigCircle' is the Vision Process.

R[5] is the status register

B.3.14 Help

This function requests IRERRHDL display help to the user 1 screen. Each time this function is called a new page is displayed. After the last page has been displayed, the first page is displayed.

Example:

CALL IRERRHDL('Help')

'Help' is the function code

B.3.15 InitMacros

This function requests IRERRHDL to load the System Macro Table. Run_Get_Offset and Run_pass_fail are loaded into the table at the entry point specified. If the macro already in the table at another location it is deleted and moved to the new location.

Example:

```
CALL IRERRHDL('InitMacros',0,10)
```

'Initmacros' is the function code

The constant 10 is the first entry point to be used.

B.3.16 SetVar

- 1 This function requests IRERRHDL to set one of the control variables to a new value. There are five control variables. Note that the 3rd parameter is always enclosed in quotes. IRERRHDL will convert the string to a Boolean, Integer, Real, or leave it as a string.

- RetryTime - Time to wait between retries, 200 milliseconds is the default
- MaxRetries - Number of times to repeat vision operation before declaring an error
- NoPartOK - 'False' is default, no part causes an error to be posted when false.
- PostNoPass - 'True' is the default value. When True an error is not logged if the test fails.
- Help - When 'Help' is the 2nd parameter, help about the variables is logged to the user page.

Example1:

```
CALL IRERRHDL('SetVar','RetryTime','250')
```

The control var RetryTime is reset to 250 milliseconds

Example2:

```
CALL IRERRHDL('SetVar','MaxRetries','1')
```

The control var MaxRetries is reset to 1. There will be no retries. The first failure is logged as an error not a warning. (0 is equivalent to 1). Values outside of 0 – 10 are reset to default of 2.

Example3:

```
CALL IRERRHDL('SetVar','NoPartOK','True')
```

The control var No Part OK is reset to True.

Example4:

```
CALL IRERRHDL('SetVar','PostNoPass','False')
```

The control var PostNoPass is set to False. When a Pass Fail request fails or is inconclusive, the event is posted.

Example5:

```
CALL IRERRHDL('SetVar','Hel'p)
```

All the Control vars are displayed with help to the user 1 page.